

DATE: EXPT NO:08	MINI PROJECT Airline Passenger Satisfaction Analytics
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AIM:

To analyze airline passenger satisfaction data using Python, SQL Server, and Tableau by cleaning and managing passenger records, performing data analysis, and creating interactive visualizations to identify factors affecting customer satisfaction such as travel class, service quality, flight distance, delays, and onboard facilities.

Software and Tools Required

- Python
- VS Code
- Pandas
- SQL Server
- SQL Server Management Studio (SSMS)
- Tableau Desktop

3. Process / Procedure

Step 1 — Project Setup

Create a project folder:

AirlinePassengerAnalysis

|

|— .venv

|— analysis.py

|— insert_data.py

|— airline_passenger_satisfaction.csv

Create and activate the Python virtual environment.

Step 2 — Create Database

Open SQL Server Management Studio (SSMS) and connect using:

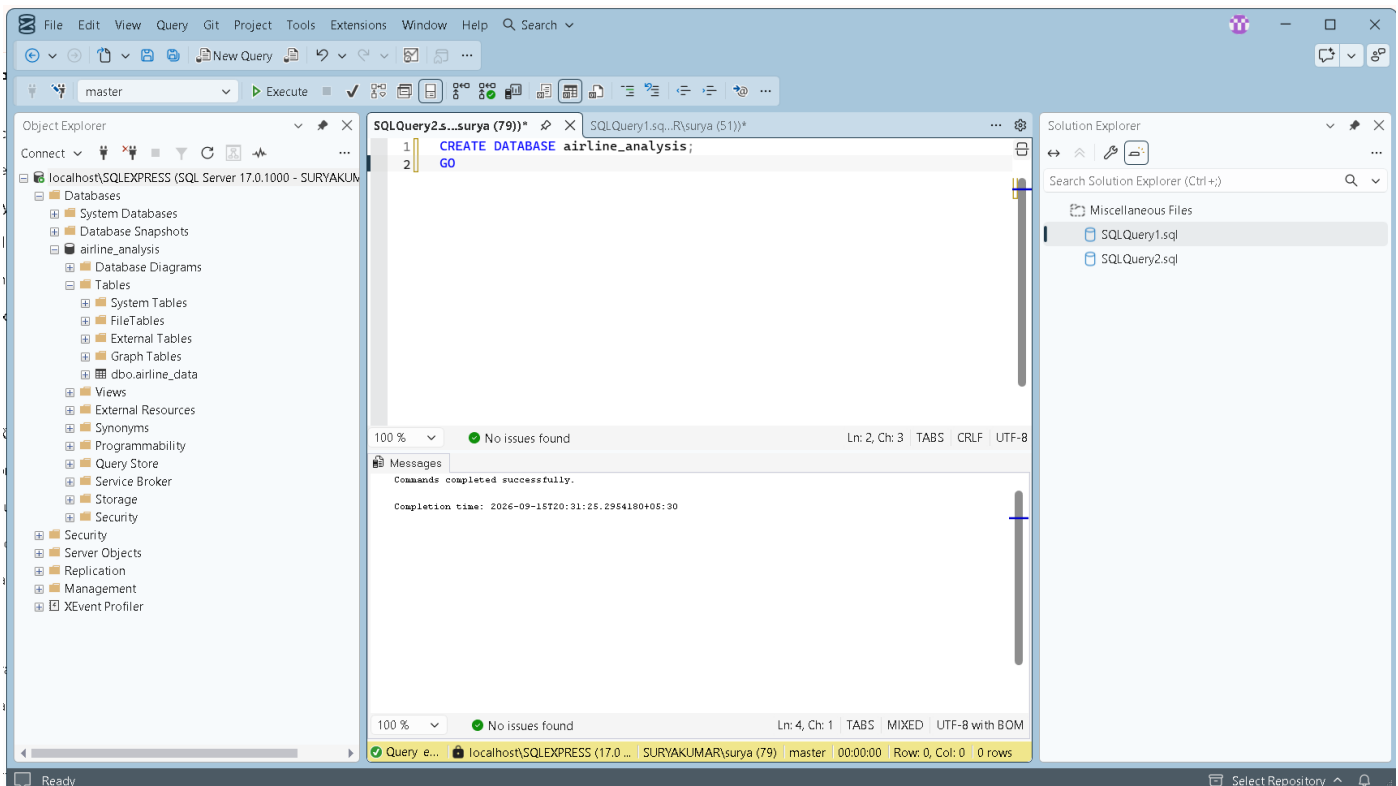
Server: localhost

Authentication: Windows Authentication

Create the database:

```
CREATE DATABASE airline_analysis;  
GO
```

OUTPUT:



Step 3 — Create Table

Create the **airline_data** table:

```
USE airline_analysis;  
GO
```

```
CREATE TABLE airline_data (  
    Passenger_ID INT,  
    Gender VARCHAR(20),  
    Customer_Type VARCHAR(50),  
    Age INT,
```

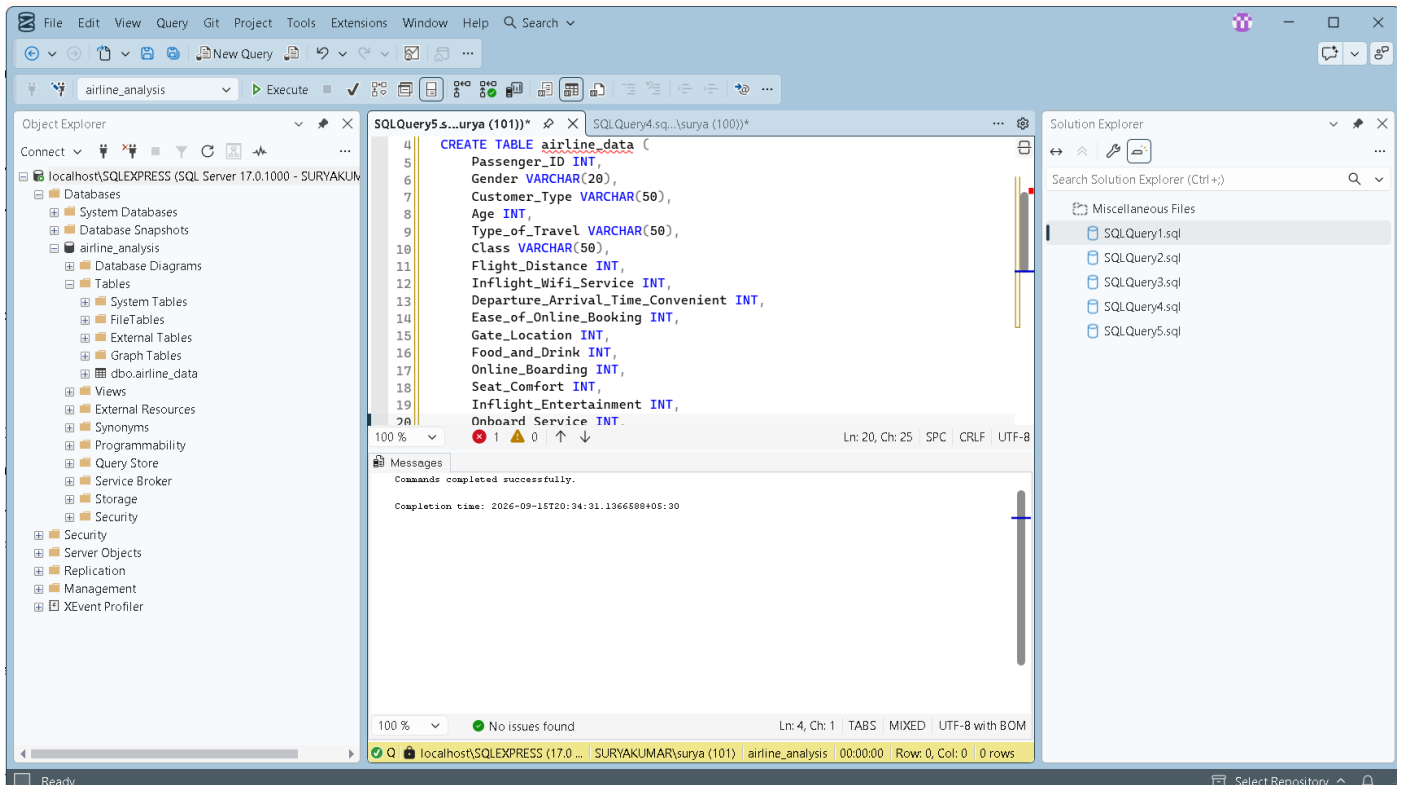
**Type_of_Travel VARCHAR(50),
Class VARCHAR(50),
Flight_Distance INT,**

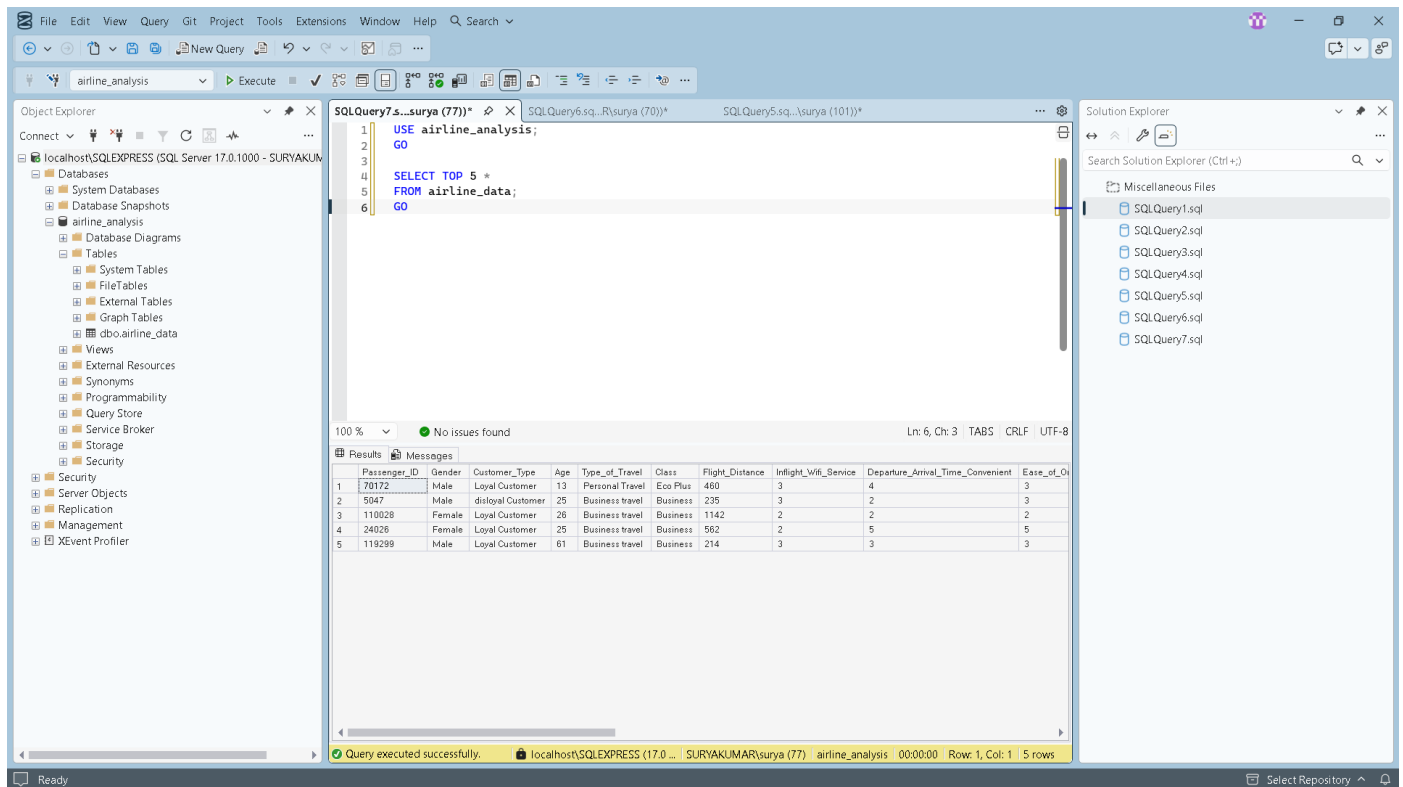
**Inflight_Wifi_Service INT,
Departure_Arrival_Time_Convenient INT,
Ease_of_Online_Booking INT,
Gate_Location INT,
Food_and_Drink INT,
Online_Boarding INT,
Seat_Comfort INT,
Inflight_Entertainment INT,
Onboard_Service INT,
Leg_Room_Service INT,
Baggage_Handling INT,
Checkin_Service INT,
Inflight_Service INT,
Cleanliness INT,
Departure_Delay INT,
Arrival_Delay FLOAT,
Satisfaction VARCHAR(50)**

);

GO

OUTPUT:





Step 4 — Load and Analyze Dataset Using Python

Program 1 — Load Dataset and View Information

Code

```
import pandas as pd

# Load dataset
df = pd.read_csv("airline_passenger_satisfaction.csv")

# Show first 5 rows
print("FIRST 5 ROWS:")
print(df.head())

print("\n")

# Dataset information
print("DATASET INFO:")
print(df.info())
```

Purpose

Load the dataset and display the first five records, column names, data types, and dataset information.

Output

```
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> & C:\Python314\python.exe c:/Users/surya/OneDrive/Desktop/AirlinePassengerAnalysis/analysis.py
FIRST 5 ROWS:
   Unnamed: 0  ...      satisfaction
0           0  ...  neutral or dissatisfied
1           1  ...  neutral or dissatisfied
2           2  ...      satisfied
3           3  ...  neutral or dissatisfied
4           4  ...      satisfied

[5 rows x 25 columns]
```

Program 2 — Check Missing Values

Code

```
import pandas as pd

# Load dataset
df = pd.read_csv("airline_passenger_satisfaction.csv")

print("MISSING VALUES:")
print(df.isnull().sum())
```

Purpose

Check for missing values in each column of the dataset.

Output

```
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> & C:\Python314\python.exe c:/Users/surya/OneDrive/Desktop/AirlinePassengerAnalysis/analysis.py
MISSING VALUES:
   Unnamed: 0      0
   id         0
   Gender      0
   Customer Type  0
   Age         0
   Type of Travel  0
   Class       0
   Flight Distance  0
   Inflight wifi service  0
   Departure/Arrival time convenient  0
   Ease of Online booking  0
   Gate location      0
   Food and drink     0
   Online boarding    0
   Seat comfort       0
   Inflight entertainment  0
   On-board service   0
   Leg room service   0
   Baggage handling   0
   Checkin service    0
   Inflight service   0
   Cleanliness        0
   Departure Delay in Minutes  0
   Arrival Delay in Minutes    310
   satisfaction        0
dtype: int64
```

Program 3 — Data Cleaning

Code

```
import pandas as pd

# Load dataset
df = pd.read_csv("airline_passenger_satisfaction.csv")

print("Before Cleaning:")
print(df["Arrival Delay in Minutes"].isnull().sum())

# Fill missing values with mean
df["Arrival Delay in Minutes"] = df["Arrival Delay in Minutes"].fillna(
    df["Arrival Delay in Minutes"].mean()
)

print("\nAfter Cleaning:")
print(df["Arrival Delay in Minutes"].isnull().sum())

# Save cleaned dataset
df.to_csv("cleaned_airline_data.csv", index=False)

print("\nCleaned dataset saved successfully!")
```

Purpose

Replace missing values in "Arrival Delay in Minutes" with the mean value and save the cleaned dataset.

Output

```
C:\Users\surya> cd C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis & python c:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis\analysis.py
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> & C:\Python314\python.exe c:/Users/surya/OneDrive/Desktop/AirlinePassengerAnalysis/analysis.py
Before Cleaning:
310

After Cleaning:
0
```

Summary

Dataset Name: Airline Passenger Satisfaction Dataset

Total Records: 103,904

Total Columns: 25

Missing Values Found: 310

Missing Values After Cleaning: 0

Output File Created: cleaned_airline_data.csv

Step 5 — Insert Data into SQL Server

The Python `insert_data.py` program connects to SQL Server and inserts the cleaned dataset into the database.

The data flow is:

Python

↓

`cleaned_airline_data.csv`

↓

SQL Server

↓

`airline_analysis`

↓

`airline_data`

CODE:

```
import pandas as pd
import pyodbc

# Connect to SQL Server
conn = pyodbc.connect(
    r"DRIVER={SQL Server};"
    r"SERVER=localhost\SQLEXPRESS;"
    r"DATABASE=airline_analysis;"
    r"Trusted_Connection=yes;"
)

cursor = conn.cursor()

# Load cleaned CSV
df = pd.read_csv("cleaned_airline_data.csv")

# Insert rows
for index, row in df.iterrows():

    cursor.execute("""
INSERT INTO airline_data VALUES (
    ?,?,?,?,?,?,?,?,
    ?,?,?,?,?,?,?,?,
    ?,?,?,?
    )
    """)
```

```

"""
    int(row['id']),
    row['Gender'],
    row['Customer Type'],
    int(row['Age']),
    row['Type of Travel'],
    row['Class'],
    int(row['Flight Distance']),
    int(row['Inflight wifi service']),
    int(row['Departure/Arrival time convenient']),
    int(row['Ease of Online booking']),
    int(row['Gate location']),
    int(row['Food and drink']),
    int(row['Online boarding']),
    int(row['Seat comfort']),
    int(row['Inflight entertainment']),
    int(row['On-board service']),
    int(row['Leg room service']),
    int(row['Baggage handling']),
    int(row['Checkin service']),
    int(row['Inflight service']),
    int(row['Cleanliness']),
    int(row['Departure Delay in Minutes']),
    float(row['Arrival Delay in Minutes']),
    row['satisfaction']
)

conn.commit()

print("Data inserted successfully!")

cursor.close()
conn.close()

```

OUTPUT:

```

PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> pip install pyodbc
Defaulting to user installation because normal site-packages is not writeable
Collecting pyodbc
  Downloading pyodbc-5.3.0-cp314-cp314-win_amd64.whl.metadata (2.8 kB)
  Downloading pyodbc-5.3.0-cp314-cp314-win_amd64.whl (72 kB)
Installing collected packages: pyodbc
Successfully installed pyodbc-5.3.0
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> python insert_data.py
Data inserted successfully!
PS C:\Users\surya\OneDrive\Desktop\AirlinePassengerAnalysis> 

```

Step 6 — Verify the Dataset

Run:

**USE airline_analysis;
GO**

**SELECT COUNT(*) AS total_records
FROM airline_data;**

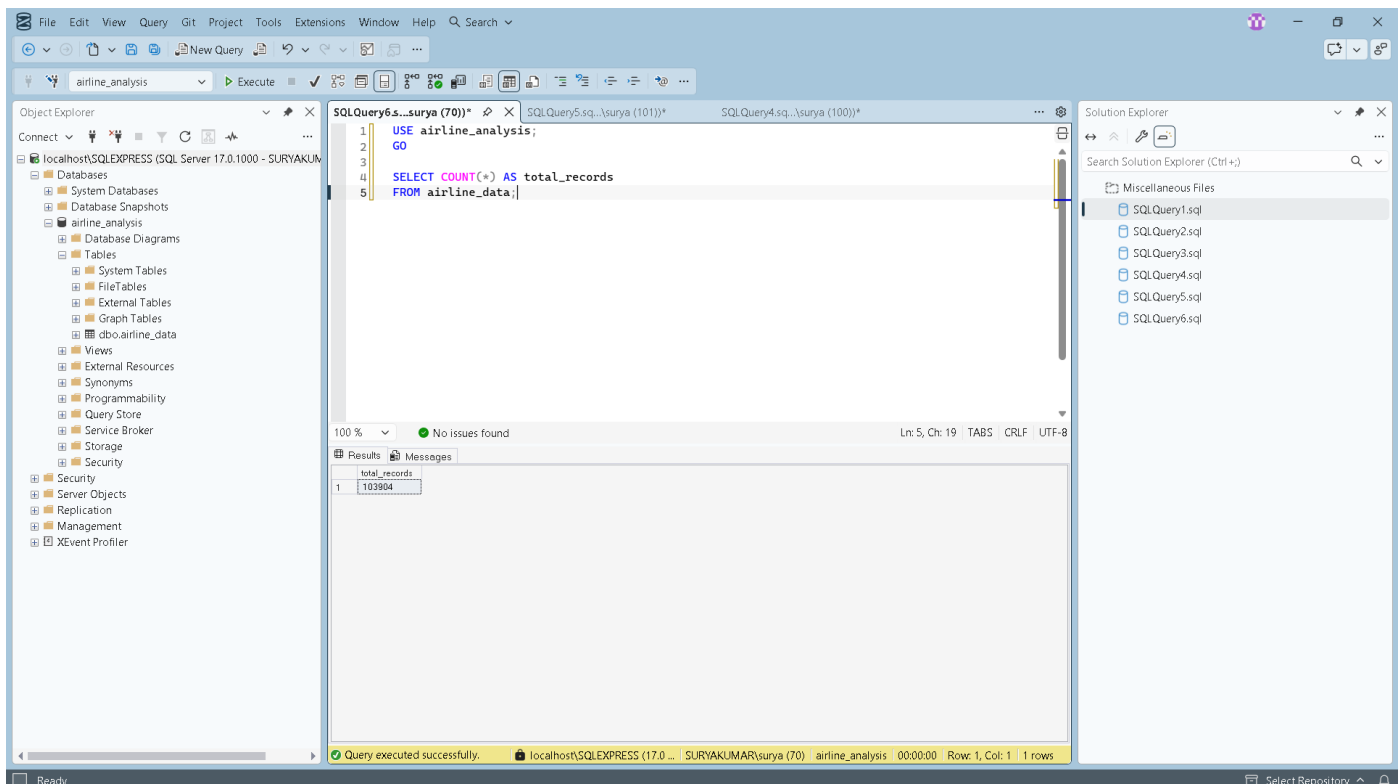
Result:

total_records

103904

This confirms that all passenger records have been successfully stored in SQL Server.

OUTPUT



3. Data Visualization Using Power BI

After storing the dataset in SQL Server, connect Power BI Desktop to:

SQL Server



airline_analysis



airline_data

Create different visualizations to analyze passenger satisfaction and airline service quality.

Visualization 1 — Satisfaction Distribution

Chart: Pie Chart

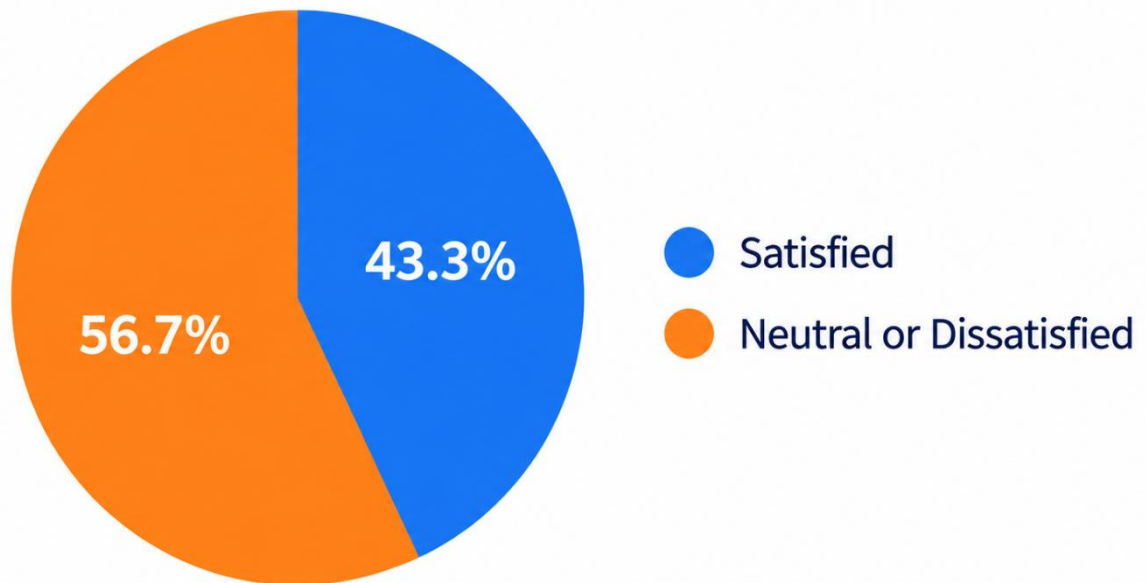
Shows the distribution of passengers based on satisfaction level.

Satisfied

Neutral or Dissatisfied

Purpose: To identify the overall passenger satisfaction percentage.

Passenger Satisfaction Distribution



Visualization 2 — Class-wise Satisfaction Analysis

Chart: Bar Chart

Shows passenger satisfaction across different travel classes.

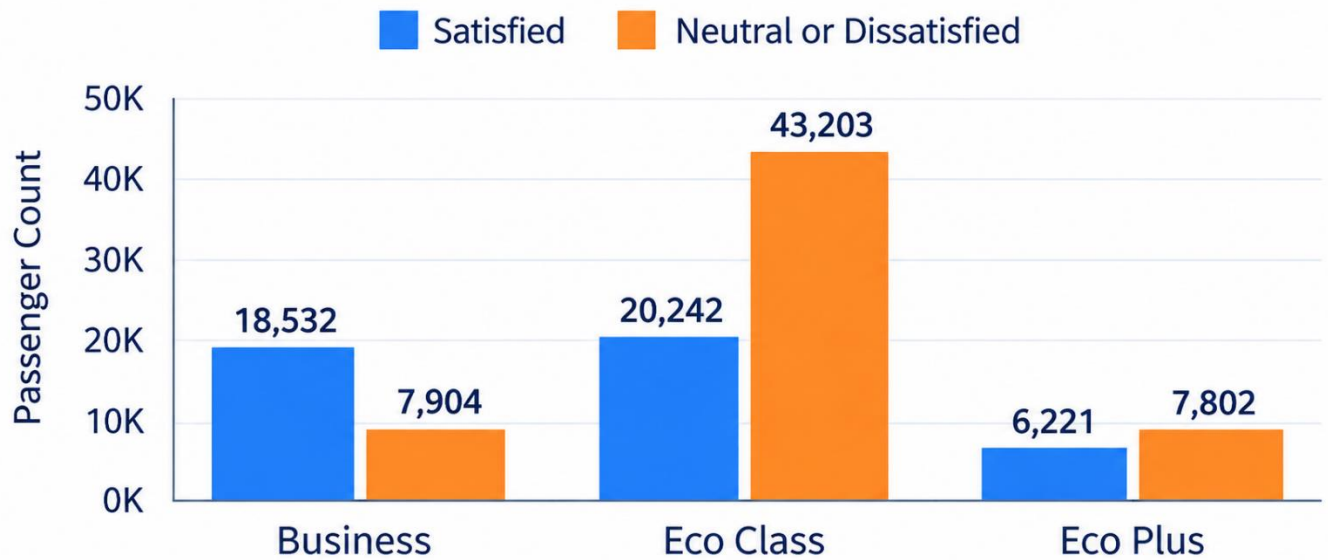
Business

Eco

Eco Plus

Purpose: To analyze how travel class affects passenger satisfaction.

Class-wise Satisfaction Analysis



Visualization 3 — Gender Distribution

Chart: Pie Chart

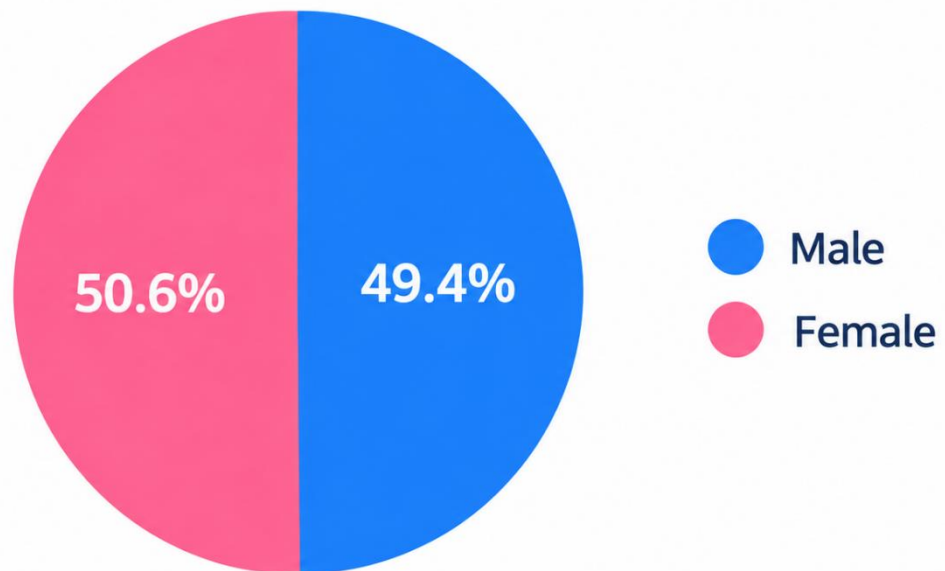
Shows the distribution of passengers by gender.

Male

Female

Purpose: To understand the passenger demographic distribution.

Gender Distribution



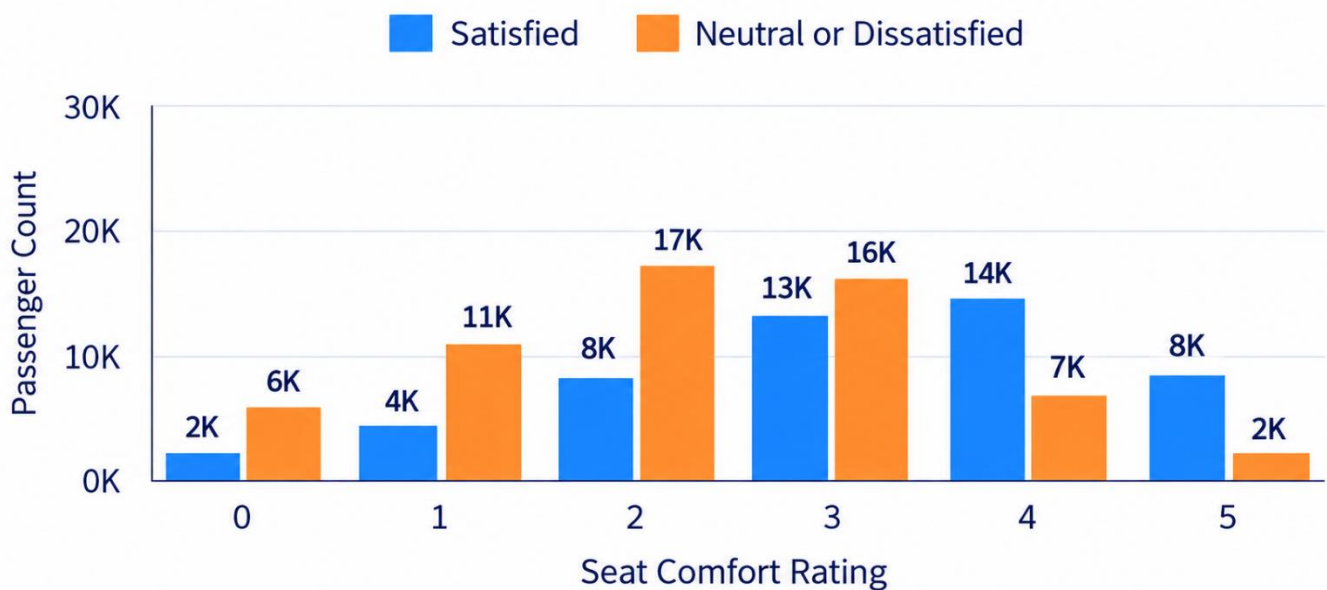
Visualization 4 — Seat Comfort vs Satisfaction

Chart: Bar Chart

Compares average seat comfort ratings with passenger satisfaction levels.

Purpose: To determine the impact of seat comfort on customer satisfaction.

Seat Comfort vs Satisfaction



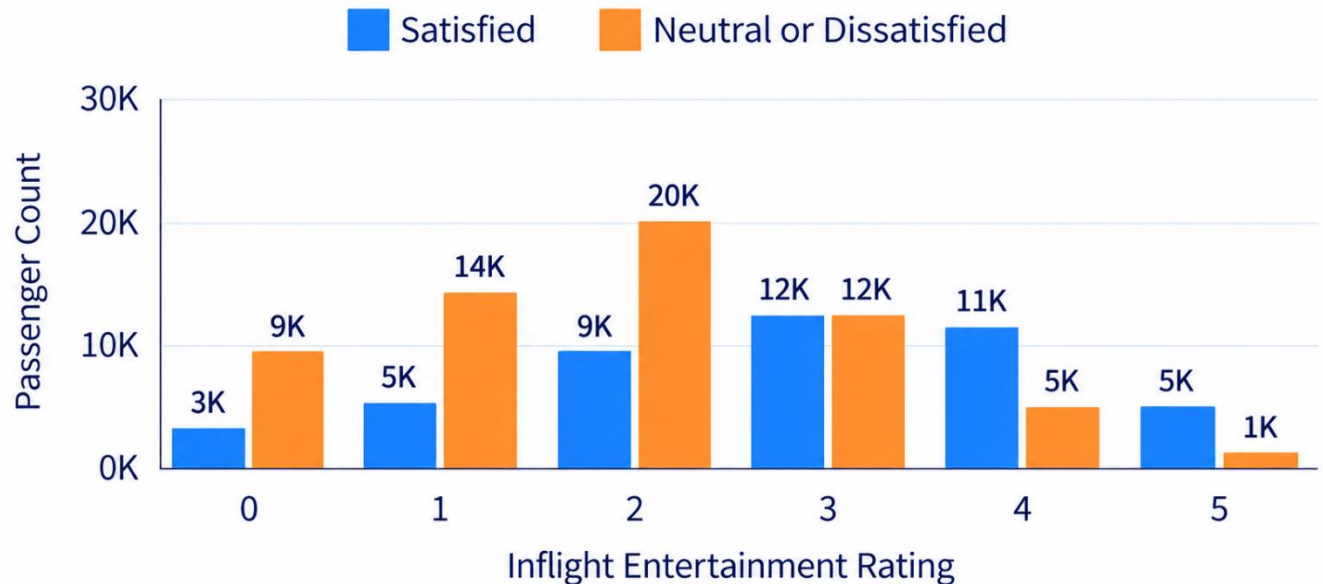
Visualization 5 — Inflight Entertainment vs Satisfaction

Chart: Bar Chart

Compares inflight entertainment ratings with passenger satisfaction levels.

Purpose: To determine the impact of inflight entertainment quality on customer satisfaction.

Inflight Entertainment vs Satisfaction



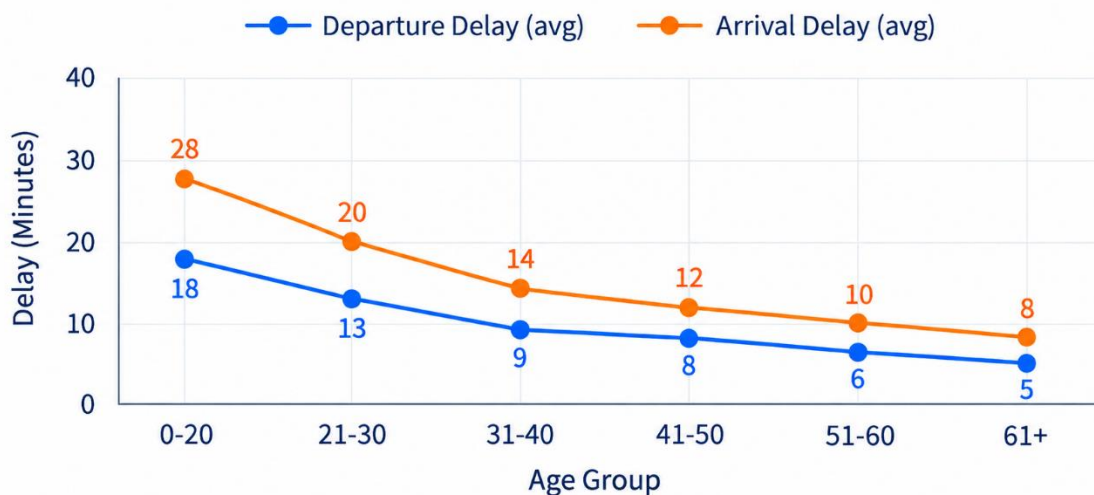
Visualization 6 — Departure and Arrival Delay Analysis

Chart: Line Chart

Compares average departure and arrival delays across different age groups.

Purpose: To analyze delay patterns among passenger age groups and identify whether certain age groups experience higher average travel delays.

Departure and Arrival Delay Analysis



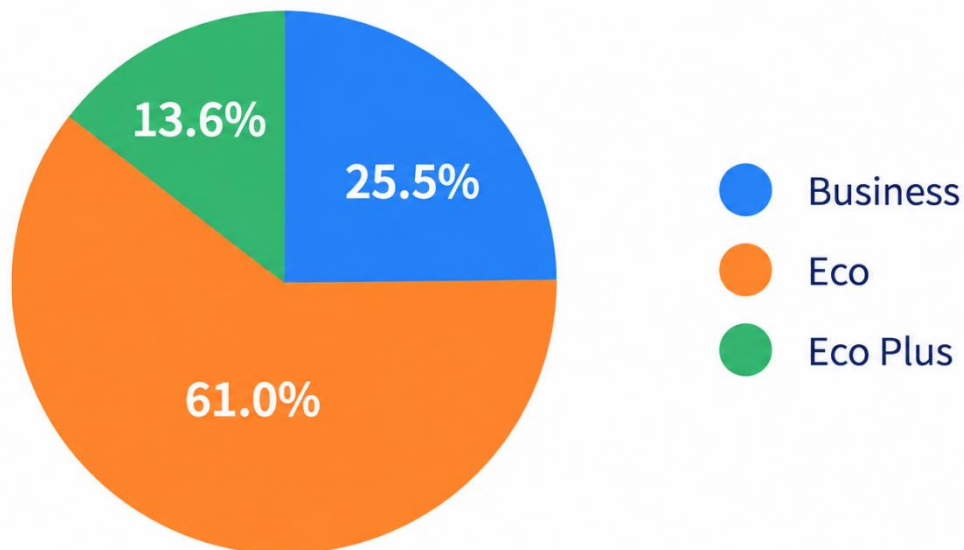
Visualization 7 — Passenger Distribution by Travel Class

Chart: Pie Chart

Shows the percentage distribution of passengers across different travel classes.

Purpose: To analyze the proportion of passengers traveling in Business, Eco, and Eco Plus classes.

Passenger Distribution by Travel Class



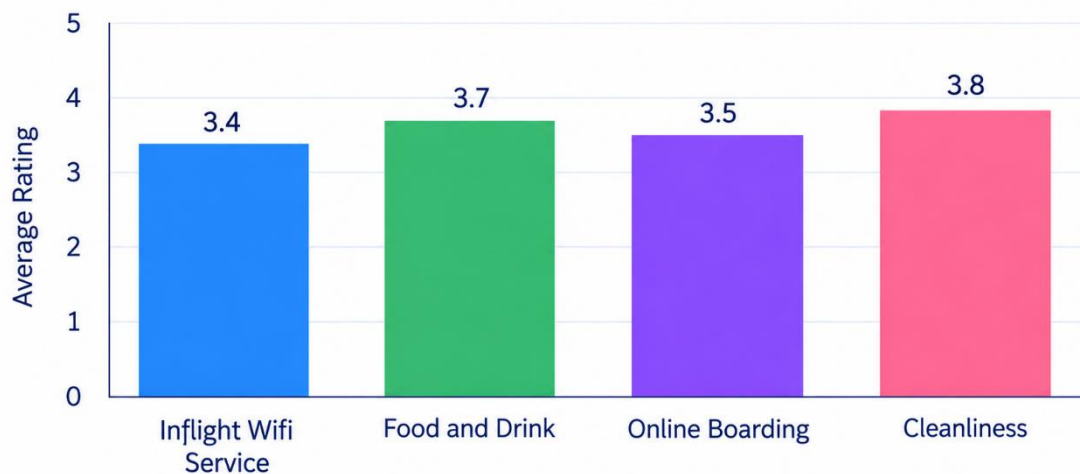
Visualization 8 — Service Quality Analysis (Average Ratings)

Chart: Bar Chart

Compares average ratings across key airline service quality factors.

Purpose: To evaluate passenger perceptions of different service aspects and identify areas with higher or lower satisfaction levels.

Service Quality Analysis (Average Ratings)



Dashboard

Combine all visualizations into a single interactive Power BI Dashboard.

The dashboard can include filters for:

Gender

Class

Customer Type

Type of Travel

Satisfaction



RESULT:

The Airline Passenger Satisfaction Analysis mini project was successfully implemented using Python, SQL Server, and Power BI. The airline passenger dataset containing 103,904 records was imported, cleaned, and analyzed using Python. Missing values were identified and handled to improve data quality. The cleaned dataset was then stored in SQL Server for efficient data management and querying. The data was successfully visualized using Power BI through bar charts, pie charts, line charts, and interactive dashboard components. The dashboard provides insights into passenger demographics, travel patterns, service quality ratings, flight delays, and customer satisfaction levels.

Final Result

103,904 Records Imported

↓

Python Data Cleaning & Analysis

↓

Cleaned CSV Dataset Created

↓

SQL Server Database Storage

↓

103,904 Records Verified

↓

Power BI Connection

↓

Charts & Visualizations Created

↓

Interactive Dashboard Developed

↓

Airline Passenger Satisfaction Analysis Completed

The project successfully demonstrates how data analytics, database management, and business intelligence tools can be used together to analyze passenger experiences and identify factors affecting customer satisfaction in the airline industry.